

```

HALF SUBTRACTOR CODE=library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

---- Uncomment the following library declaration if instantiating
---- any Xilinx primitives in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity half_subtractor is
Port ( A : in STD_LOGIC;
      B : in STD_LOGIC;
      DIFFERENCE : out STD_LOGIC;
      BORROW : out STD_LOGIC);
end half_subtractor;

architecture Behavioral of half_subtractor is

begin
DIFFERENCE <= A xor B;
BORROW <= (not A) and B;

end Behavioral;

```

```

FULL SUBTRACTOR CODE= library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

---- Uncomment the following library declaration if instantiating
---- any Xilinx primitives in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity full_subtractor is
Port ( A : in STD_LOGIC;
      B : in STD_LOGIC;
      Bin : in STD_LOGIC;
      DIFFERENCE : out STD_LOGIC;
      BORROW : out STD_LOGIC);
end full_adder;

```

architecture Behavioral of full_adder is

SIGNAL D1,B1,B2:STD_LOGIC;

begin

half_subtractor1:ENTITY WORK.half_subtractor PORT MAP(A,B,D1,B1);

half_subtractor2:ENTITY WORK.half_subtractor PORT MAP(D1,Bin,DIFFERENCE,B2);

BORROW<=B1 or B2;

end Behavioral;